16 An ideal gas is held in a container of volume V at pressure P. The root-mean-square (r.m.s) speed of a gas molecule under these conditions is u. If the volume and pressure are now changed to $2V$ and $2P$ respectively, what will be the r.m.s speed of a molecule?

A u **B** $\sqrt{2}u$ **C** $2u$ **D** $4u$